72 Lumbar Microdiskectomy

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INDICATIONS

- Patients with back pain, sciatic pain, Lasègue sign (pain with straight leg raise), or sensory deficit that fails to improve with conservative measures.
- Referable foraminal stenosis from posterolateral disk herniation.
- New or progressive motor deficits require more urgent surgical intervention.

CONTRAINDICATIONS

- Large disk herniation causing central canal stenosis requires a laminectomy.
- A more extensive procedure, such as interbody fusion, may be beneficial to patients with recurrent disk herniation or concomitant instability from degenerative disease or spondylolisthesis.

PLANNING AND POSITIONING

- Preoperative evaluation includes a thorough neurologic history and sensory, motor, and reflex examination. The sensory examination should be conducted to rule out any dermatomal distribution of sensory loss. Routine lateral, flexion, and extension lumbar spine films can show dynamic instability that may necessitate additional lumbar fusion. In the event of instability, a diskectomy would treat the radiculopathic leg pain but would not improve mechanical back pain. Magnetic resonance imaging (MRI) without contrast enhancement of the lumbar spine can show focal neural foraminal stenosis and disk herniation and allows selection of the appropriate level for diskectomy and foraminotomy.
- Either prone or lateral position is acceptable, depending on surgeon preference.

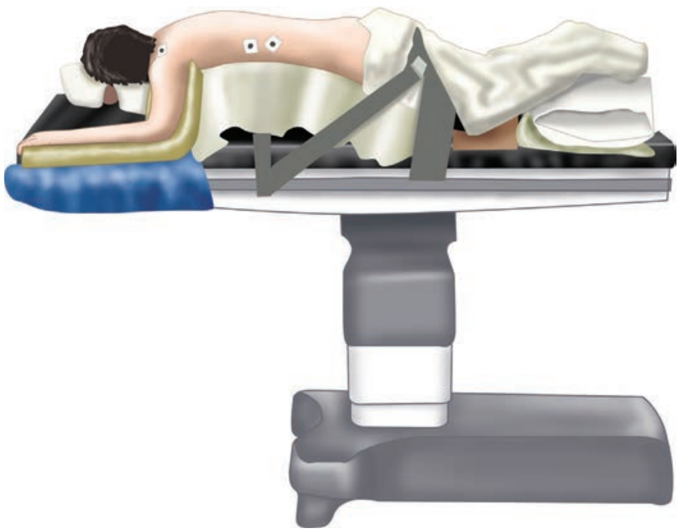

Fig. 72.1 The patient is placed prone on chest rolls or a Wilson frame to hold the spine in flexion.

- The patient should be given a dose of preoperative antibiotics prior to skin incision.

PROCEDURE

Skin Incision

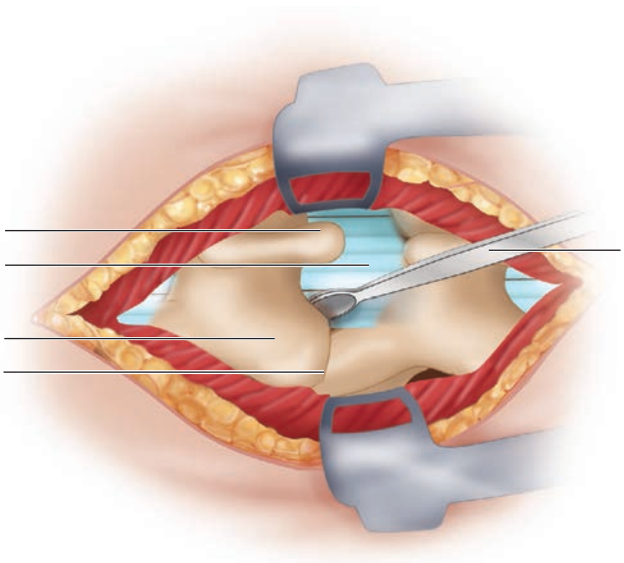

Fig. 72.2 By palpating the anterior superior iliac crest, the L4-5 interspace can be localized for a rough estimation of the level. A spinal needle localization film helps determine the correct site for the exposure. After injection of local anesthetic with epinephrine into the skin and deeper tissues, a small incision just lateral to the midline should be made using a No. 10 blade.

Fascial and Subfascial Dissection

- Bovie electrocautery can be used for subcutaneous dissection and for achieving hemostasis. The subcutaneous fat is dissected until the thoracolumbar fascia is identifiable. A dry sponge raked along the fascia can identify the white tissue easily, and a periosteal elevator can be used to dissect off the subcutaneous fat from the fascia beneath.

Subperiosteal Dissection

- Bovie electrocautery on cut function can be used to incise the fascia. For microdiskectomy, a paramedian facial incision helps avoid damaging midline ligamentous structures. After a minimal, one-sided exposure is completed, a localizing film should be done to confirm the correct level.
- Subperiosteal dissection can also be done rapidly with an open dry sponge raked ventrally and laterally along the spinous process and lamina with a large periosteal dissector, such as a Cobb elevator.